National Neighborhood Data Archive (NaNDA): Eating and Drinking Places by Census Tract, United States, 2003-2017

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Overview and Data Dictionary

Documentation Version: 1.0
Last updated: 9/16/2020

Dataset Overview

Description

This dataset contains measures of the number and density of eating and drinking places – including fast food restaurants, coffee shops, and bars – per United States census tract from 2003 through 2017.

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Funding Sources

- United States Department of Health and Human Services. Administration for Community Living. National Institute on Disability, Independent Living, and Rehabilitation Research (90RTHF0001)
- United States Department of Health and Human Services. National Institutes of Health. National Institute on Aging (RF1-AG-057540)

Data Sources

The North American Industry Classification System (NAICS) was used to identify organizations that can be considered eating and drinking places. NAICS was developed by the United States Office of Management and Budget and is “the standard used by Federal statistical agencies in classifying business establishments for the purpose of collecting, analyzing, and publishing statistical data related to the U.S. business economy” (United States Census Bureau, 2017).

Establishment data was taken from the National Establishment Time Series (NETS) database, which provides annual records of the US economy for more than 60 million private for-profit and nonprofit establishments and government agencies (Walls & Associates, 2017). This longitudinal data source classifies establishments by NAICS code and provides detailed business microdata and address history. NETS is produced in most years by Walls and Associates using archival establishment data from Dun & Bradstreet. While it currently includes data from 1990 to 2015 and 2017, only data from 2003-2015 and 2017 were used to create this dataset. (NETS data was not produced for 2016.)

Census tracts for establishments were determined using the 2010 TIGER/Line shapefiles produced by the United States Census Bureau (United States Census Bureau, 2010).

Census tract population figures for 2000-2009 were taken from Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019). Population figures for 2010-2017 were obtained from Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020).

Coverage

The dataset contains one observation per census tract per year in the United States, Puerto Rico, and the US Virgin Islands. Other US island territories are excluded.

Methodology

To produce this dataset, we defined several types of organizations that function as common “third places” in the United States based on qualitative research conducted by members of our research team (Finlay et al., 2018). Third places are “physical locations outside of the home (first place) or workplace (second place) that facilitate social interaction, community building, and social support" (Finlay et al., 2019).

One such type of place is eating and drinking places. Prior research has demonstrated the importance of fast food restaurants, coffee shops, and bars as social gathering places (Finlay et al., 2019, Broughton et al., 2016, Rosenbaum et al., 2007). Proximity to high densities of eating and drinking places may play a role in maintaining cognitive function among older adults (Finlay et al., 2020). Total expenditures on food away from home (in fast-food and full-service restaurants) are close to or higher than expenditures on food at home (Elitzak & Okrent, 2018), suggesting that these establishments are common destinations in daily life.

Our research team identified the following NAICS codes as representing eating and drinking places:

- 7225: all restaurants and eating places.
- 722511: full-service restaurants where patrons order and are served while seated.
- 722513: limited service restaurants, e.g. fast-food restaurants.
- 722515: snack and nonalcoholic beverage bars, such as coffee shops, doughnut or bagel shops, and ice cream parlors.
- 722410: drinking places (alcoholic beverages), such as bars, taverns, and cocktail lounges.

Note that NAICS codes are hierarchical, and four-digit codes are inclusive of all six-digit codes that start with the same pattern. For example, if a tract contains five eating places (7225), two restaurants (722511), two fast food restaurants (722513), and one coffee shop (722515), the total number of providers is five, not ten.

For each NAICS code we identified, we queried the NETS database for all establishments that were open at some time between 2003 and 2015 and in 2017 (NETS does not contain data for 2016), and their location during each year. Note that NETS data were not cleaned or verified during this process, and that NETS data has limitations and inaccuracies as described in Gomez-Lopez et al. (2017) and Finlay et al. (2019).

We determined each establishment's census tract for each year by mapping its latitude and longitude provided in the NETS database to the 2010 version of the TIGER/Line shapefiles from the US Census Bureau.

Using these locations, we created three variables to represent establishment counts per tract. The first count variable is the total of all establishments with that NAICS code in the tract. The second is the count of just establishments with non-zero sales figures in the current year, which can be used where desired to filter out establishments with very low business volume. The third is the count of establishments employing more than one person, which can be used to filter out very small owner-operated businesses if desired. The most useful variable for a given research question may vary by establishment type, so we have created three count types for all NAICS codes so that users may make their own decision about the version that best meets their needs.

We then calculated population densities (per 1000 people) for each of the three count variables for each NAICS code. Tract populations for 2003-2009 come from the NaNDA dataset Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010. Values for each year are interpolated. For 2010-2012 and 2013-2017, we used the tract population from the NaNDA dataset Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017. Population figures in this dataset were taken from the 2012 and 2017 ACS five-year estimates, respectively, in order to avoid overlapping ACS estimates as advised by the Census Bureau, and for compatibility with other NaNDA datasets. Population figures for 2010-2017 are not interpolated. Consult the codebooks for the NaNDA datasets Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019) and Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020) for details on why these population figures were selected and how 2003-2009 population was interpolated.

We also created three variables representing the density of establishments per square mile. The area is the land area of the tract from the 2010 version of the TIGER/Line shapefiles. As a result, all area densities are standardized to 2010 census geography and are comparable across years across the entire dataset.

Population density was set to -1 for tracts with zero population in a given year. Population and area densities are missing for tracts in Puerto Rico and the US Virgin Islands, which are not included in the TIGER/Line shapefiles. Population and area densities are also missing in certain years for a few census tracts whose FIPS codes changed at some point between 2003 and 2017.

Finally, we merged counts and densities of all selected NAICS codes to create an aggregate dataset containing tract population, area, and count and density variables for all eating and drinking places per tract in each year.

Variables

For additional information on the definition of each NAICS code, see the Methodology section of this document and the 2017 NAICS website: https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017.

Variable	Type	Obs	Unique	Mean	Min	Max	Label
tract_fips10	string	1036462	74033	.	.	.	Census tract FIPS code (2010 TIGER/Line shapefiles)
year	int	1036462	14	2009.571	2003	2017	Year
population	float	1020881	250271	4208.894	0	65528	Tract population
aland10	float	1022434	72534	48.07265	0	85425.73	Tract land area, square miles
count_7225	int	1036462	254	8.858368	0	311	# all eating places
count_sales_7225	int	1036462	241	8.210534	0	311	# all eating places w/ sales>0
count_emp_7225	int	1036462	240	8.152124	0	296	# all eating places w/ 2+ employees
popden_7225	float	1016311	431551	94.67494	-1	3.50E+07	# all eating places per 1000 people
popden_sales_7225	float	1016282	423669	87.46504	-1	3.50E+07	# all eating places per 1000 people w/ sales>0
popden_emp_7225	float	1016272	420787	94.07275	-1	3.50E+07	# all eating places per 1000 people w/ 2+ employees
aden_7225	float	1017996	361965	13.41506	0	3432.928	# all eating places per sq mile
aden_sales_7225	float	1018004	358156	12.40028	-1	3120.844	# all eating places per sq mile w/ sales>0
aden_emp_7225	float	1018004	357483	12.2971	-1	3432.928	# all eating places per sq mile w/ 2+ employees
count_722511	int	1036462	180	5.230472	0	224	# full service restaurants
count_sales_722511	int	1036462	169	4.822695	0	224	# full service restaurants w/ sales>0
count_emp_722511	int	1036462	170	4.762698	0	214	# full service restaurants w/ 2+ employees
popden_722511	float	1016053	369276	59.35121	-1	2.34E+07	# full service restaurants per 1000 people
popden_sales_722511	float	1016027	359963	53.03399	-1	2.34E+07	# full service restaurants per 1000 people w/ sales>0
popden_emp_722511	float	1016018	356004	58.85644	-1	2.34E+07	# full service restaurants per 1000 people w/ 2+ employees
aden_722511	float	1017996	287562	8.80189	0	3432.928	# full service restaurants per sq mile
aden_sales_722511	float	1018004	283876	8.115218	-1	3120.844	# full service restaurants per sq mile w/ sales>0
aden_emp_722511	float	1018004	281388	8.015516	-1	3432.928	# full service restaurants per sq mile w/ 2+ employees
count_722513	int	1036462	74	2.076404	0	79	# fast food restaurants

Variable	Type	Obs	Unique	Mean	Min	Max	Label
count_sales_722513	int	1036462	73	1.945009	0	77	# fast food restaurants w/ sales>0
count_emp_722513	int	1036462	72	1.968	0	76	# fast food restaurants w/ 2+ employees
popden_722513	float	1015731	250642	34.05568	-1	1.17E+07	# fast food restaurants per 1000 people
popden_sales_722513	float	1015707	243111	33.26335	-1	1.17E+07	# fast food restaurants per 1000 people w/ sales>0
popden_emp_722513	float	1015708	242790	34.01409	-1	1.17E+07	# fast food restaurants per 1000 people w/ 2+ employees
aden_722513	float	1017996	149864	2.525715	0	350.6211	# fast food restaurants per sq mile
aden_sales_722513	float	1017996	148019	2.344768	0	350.6211	# fast food restaurants per sq mile w/ sales>0
aden_emp_722513	float	1017996	146807	2.37547	0	325.5768	# fast food restaurants per sq mile w/ 2+ employees
count_722515	int	1036462	41	0.6353065	0	51	# coffee shops
count_sales_722515	int	1036462	41	0.5857967	0	51	# coffee shops w/ sales>0
count_emp_722515	int	1036462	41	0.5695993	0	51	# coffee shops w/ 2+ employees
popden_722515	float	1015460	142034	0.3843828	-1	13234.82	# coffee shops per 1000 people
popden_sales_722515	float	1015440	134803	0.3573867	-1	13234.82	# coffee shops per 1000 people w/ sales>0
popden_emp_722515	float	1015440	131119	0.3447443	-1	13234.82	# coffee shops per 1000 people w/ 2+ employees
aden_722515	float	1017996	69042	0.8487371	0	232.5929	# coffee shops per sq mile
aden_sales_722515	float	1017996	68146	0.7817421	0	232.5929	# coffee shops per sq mile w/ sales>0
aden_emp_722515	float	1017996	65492	0.7667529	0	232.5929	# coffee shops per sq mile w/ 2+ employees
count_722410	int	1036462	87	1.23291	0	128	# bars
count_sales_722410	int	1036462	80	1.13464	0	118	# bars w/ sales>0
count_emp_722410	int	1036462	87	1.151841	0	126	# bars w/ 2+ employees
popden_722410	float	1015493	200504	31.0645	-1	1.17E+07	# bars per 1000 people
popden_sales_722410	float	1015469	192640	31.01678	-1	1.17E+07	# bars per 1000 people w/ sales>0
popden_emp_722410	float	1015482	192718	31.03879	-1	1.17E+07	# bars per 1000 people w/ 2+ employees
aden_722410	float	1017996	119641	1.851226	0	531.4094	# bars per sq mile
aden_sales_722410	float	1017996	116009	1.703886	0	506.1042	# bars per sq mile w/ sales>0
aden_emp_722410	float	1017996	116322	1.753996	0	489.3233	# bars per sq mile w/ 2+ employees

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